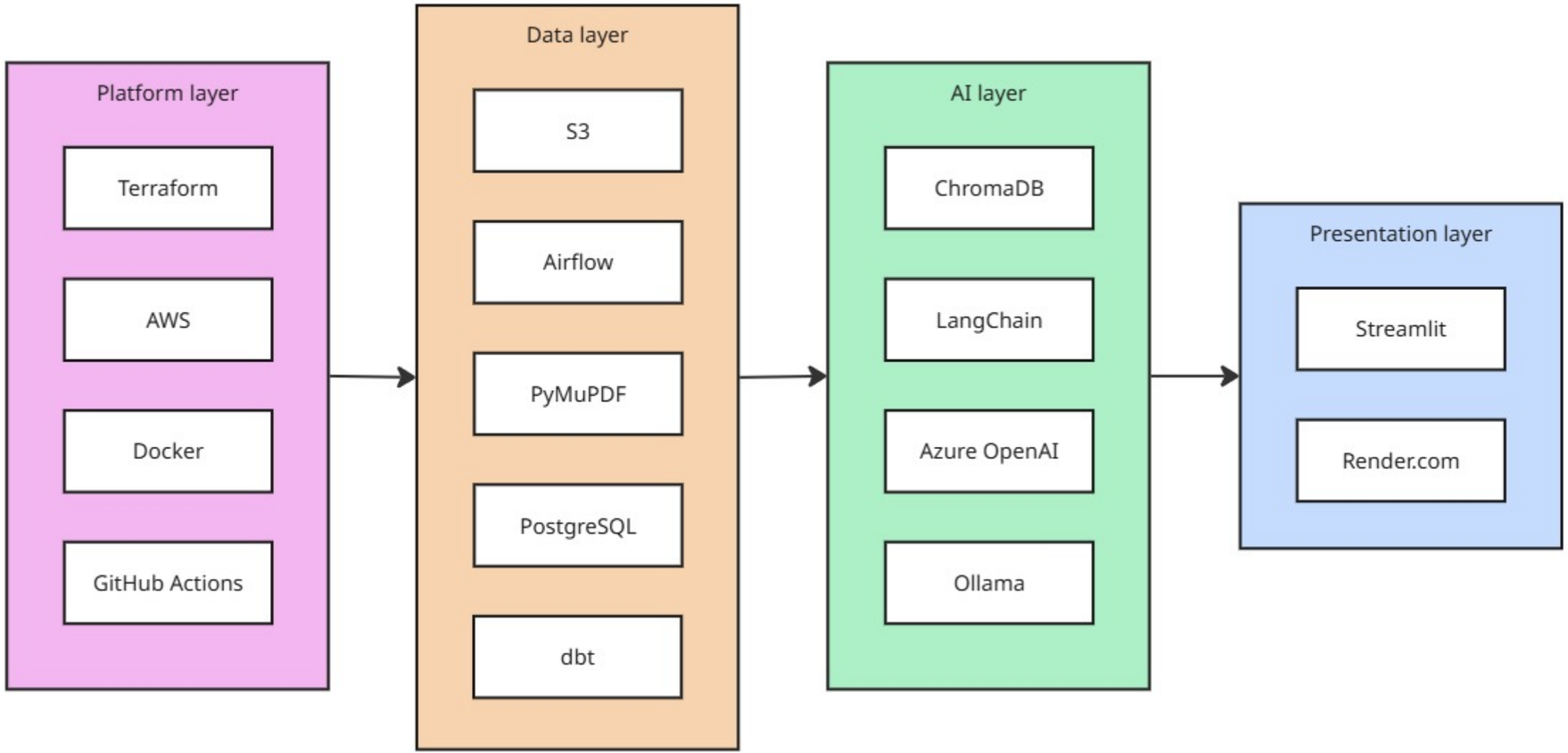
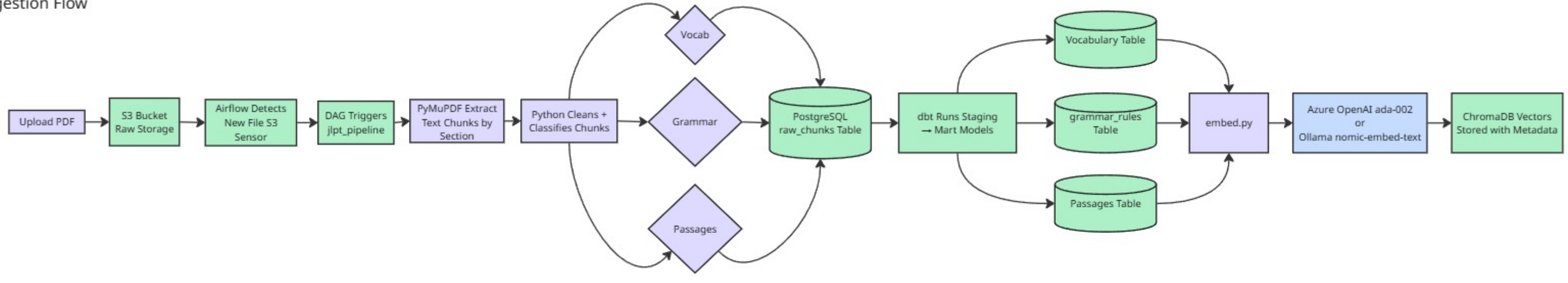
High Level System Overview



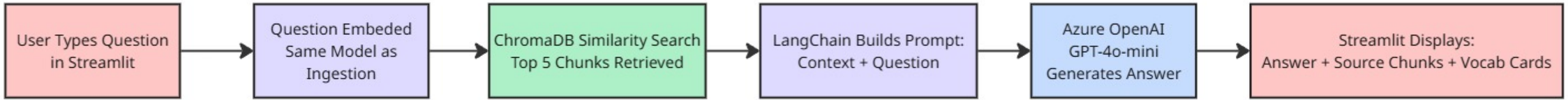
Cost Annotations:

- EC2: Free Tier — 750hrs/mo
- RDS: Free Tier — 20GB
- S3: Free Tier — 5GB (12 months)
- Azure OpenAI: Pay-as-you-go
- Streamlit/Render: Free Tier — limited hours

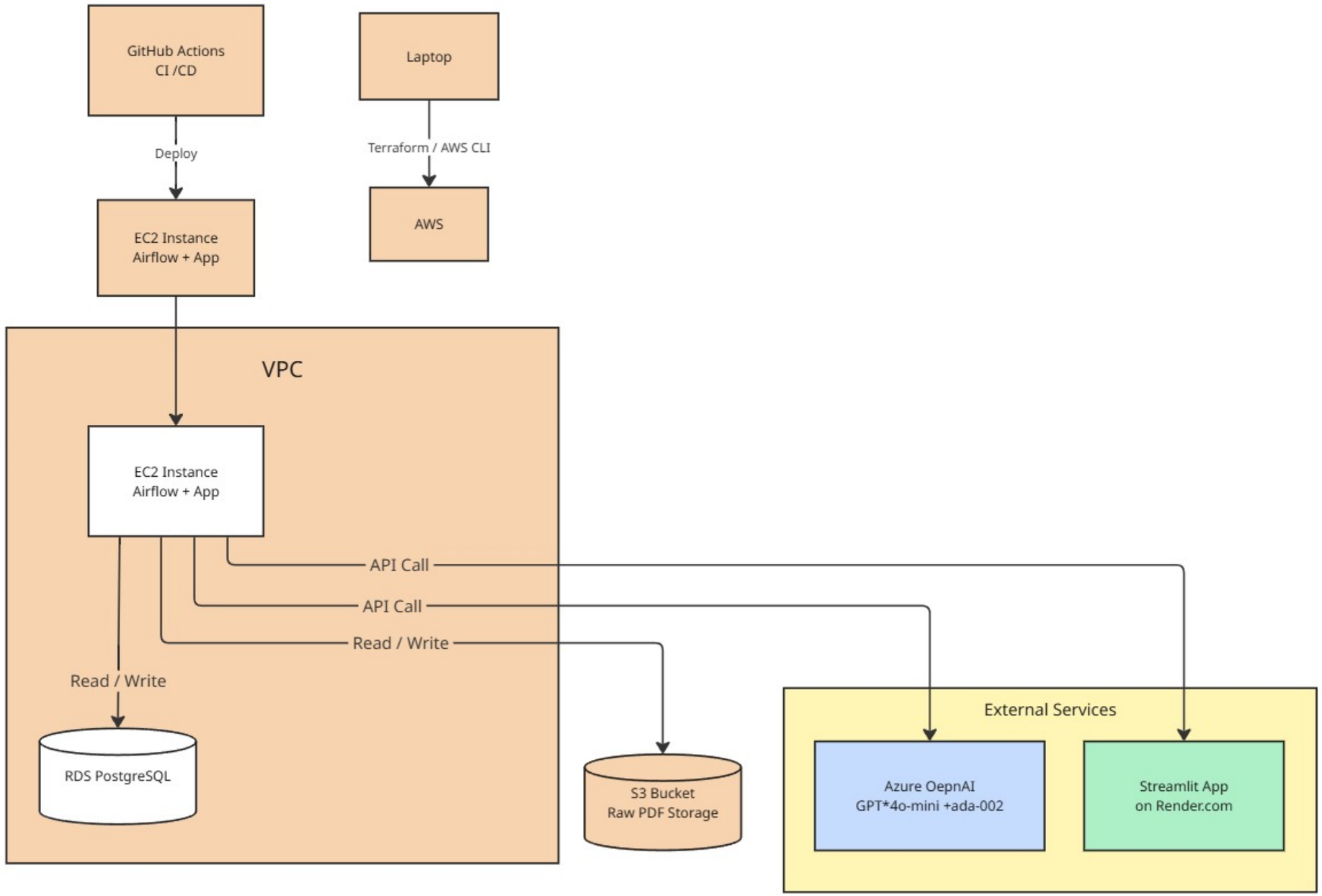
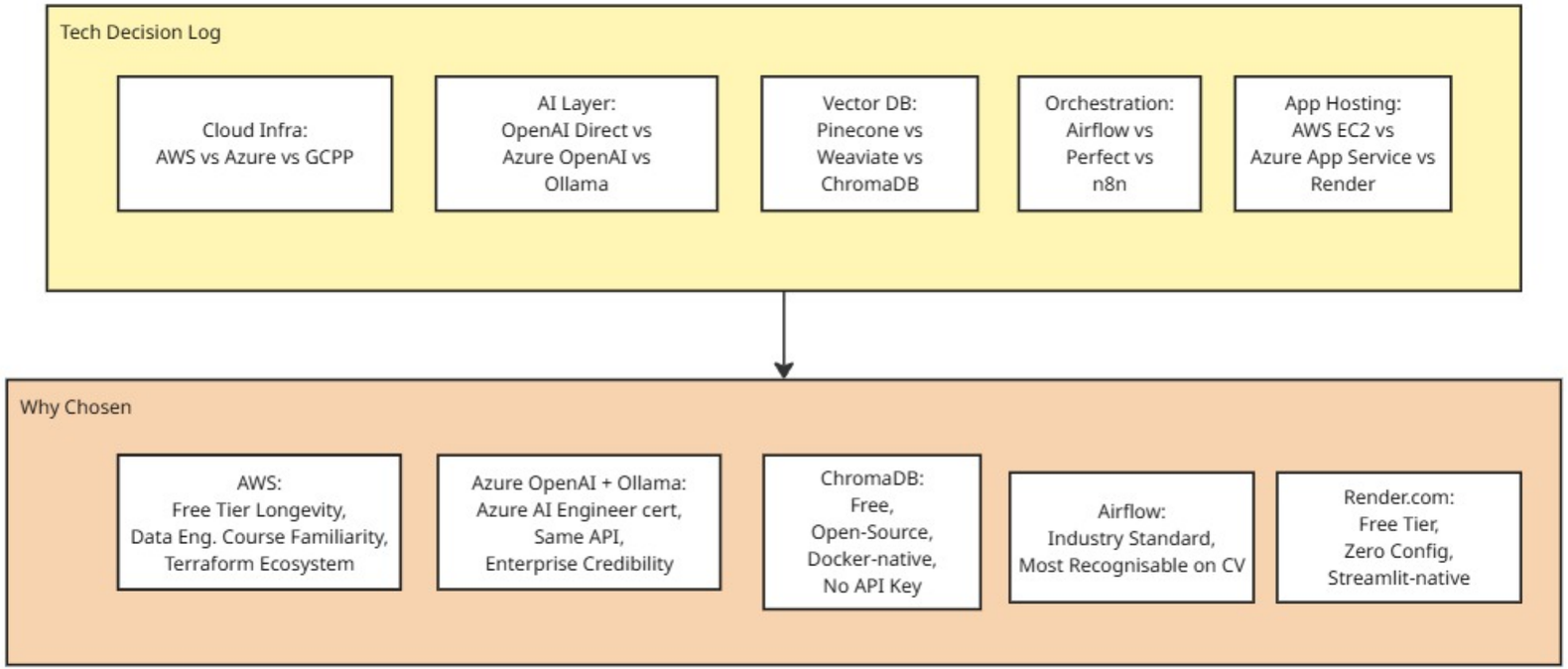
Ingestion Flow



Query Flow


User Types Question in Streamlit
Question Embeded Same Model as Ingestion
ChromaDB Similarity Search Top 5 Chunks Retrieved
LangChain Builds Prompt: Context + Question
Azure OpenAI GPT-4o-mini Generates Answer
Streamlit Displays: Answer + Source Chunks + Vocab Cards

Infrastructure Diagram (Platform Layer Detail)



Tech Decision Log